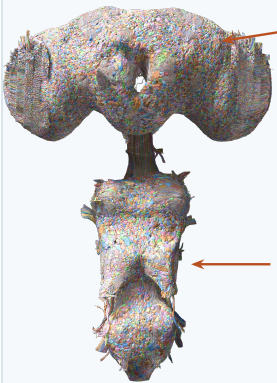
From fly anatomy to hexapod control

Anatomical reference



Brain shown for context.
The model uses a selected VNC-centered network.

Fixed network and motor map

Named descending inputs
External Poisson drive

Measured wiring

Frozen VNC model
11,000 nodes;
485,074 edges

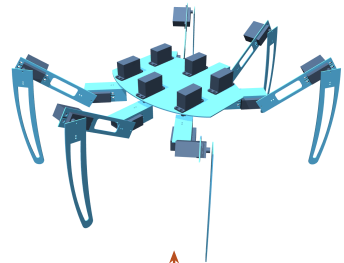
Motor rates

260 motor channels
Opposing pool means
↓
Fixed gains and joint grouping

Target angles

18 joint targets
PD joint controller
6 legs in MuJoCo

Simulated body



Fixed-map experiment: no learned conversion and no body feedback to the network.
Walking experiments add a separate rhythm generator, not shown here.

Anatomy: MaleCNS / Janelia FlyEM and collaborators, CC BY 4.0. Schematic, not to scale.